

Symmetry-forbidden intraband transitions leading to ultralow Gilbert damping in van der Waals ferromagnets

Weizhao Chen,^{1,2} Yu Zhang,¹ Yi Liu,^{3,*} and Zhe Yuan^{2,†}

¹*Center for Advanced Quantum Studies and School of Physics and Astronomy,
Beijing Normal University, Beijing 100875, China*

²*Interdisciplinary Center for Theoretical Physics and Information Sciences, Fudan University, Shanghai 200433, China*

³*Institute for Quantum Science and Technology, Department of Physics, Shanghai University, Shanghai 200444, China*
(Dated: May 28, 2025)

Based upon first-principles calculations, we report ultralow Gilbert damping in two-dimensional (2D) van der Waals (vdW) ferromagnets. The low damping occurs at low temperature because **mirror symmetry prohibits intraband transitions**. The monotonic dependence on the electronic scattering rate suggests the absent lower limit, in contrast to conventional ferromagnetic materials. Breaking mirror symmetry through magnetization rotation, layer stacking, or structural phase transition significantly **increases damping by enabling intraband transitions**. Topological nodal lines, also protected by mirror symmetry, contribute substantially to interband-transition-mediated damping, which can be tuned by adjusting the Fermi level. Our findings elucidate the unique characteristics of Gilbert damping in 2D vdW ferromagnets, providing valuable insights for designing low-dimensional spintronic devices with high energy efficiency.

Introduction.—The discovery of 2D vdW ferromagnetic (FM) materials, exemplified by CrI_3 [1], $\text{Cr}_2\text{Ge}_2\text{Te}_6$ [2], and Fe_3GeTe_2 [3, 4], has not only dramatically broadened the horizon of FM materials but also ignited promising prospects for low-dimensional spintronics devices [5, 6]. This promise stems partly from the fact that magnetism in 2D vdW FM metals can be controlled through unique approaches like gating and strain [7, 8]. As a typical 2D FM metal, Fe_3GeTe_2 has attracted considerable attention for its spin-orbit torque-driven magnetization switching [9–11], its unconventional tunability in magnetic tunnel junctions [12, 13] and its potential in novel electronic applications [14]. A very similar vdW FM metal Fe_3GaTe_2 [15] has been discovered with an almost identical lattice structure to Fe_3GeTe_2 , yet it boasts a Curie temperature above room temperature, rendering it an ideal candidate for integration into magnetic heterostructures and devices that operate under ambient conditions [16–19].

The quest for designing high-performance, low-consumption spintronics devices based on 2D vdW FM materials requires a comprehensive understanding of the dynamical properties of 2D magnetism [20], particularly the effects of reduced dimensionality. Gilbert damping, a crucial parameter in magnetization dynamics [21], determines the critical current density for current-induced magnetization switching [22], the switching time scale [23] and the velocity of magnetic texture driven by magnetic fields and/or electric currents [24, 25]. More importantly, **low Gilbert damping is essential for low-consumption in memory and logic devices**. Ultralow damping has been observed in Fe-based alloys [26–29] and half metals [30, 31]. In FM transition metals and alloys, such as Fe, Co, Ni [32–34] and FePd [35, 36], the experimentally measured damping α exhibits a non-monotonic temperature dependence: α increases with decreasing

temperature (conductivity-like) at low-temperatures and with increasing temperature (resistivity-like) at high-temperatures, limiting damping above its lower bound. However, in the recently discovered 2D FM metals, Fe_3GeTe_2 and Fe_3GaTe_2 , a monotonic temperature dependence of damping has been observed in both experimental [37] and theoretical studies [38]. Furthermore, significant damping anisotropy has been identified in these 2D FM metals [38, 39], suggesting an atypical magnetization relaxation likely due to their low-dimensional lattice structures.

In this Letter, we present **a theoretical calculation of the Gilbert damping** in 2D vdW FM metals, using Fe_3GaTe_2 as a typical example. We demonstrate that **mirror symmetry in 2D vdW systems prohibits intraband transitions, leading to the absence of conductivity-like damping**. Consequently, the damping exhibits a monotonic temperature dependence without the lower limit. An ultralow $\alpha < 10^{-3}$ is achieved in the single layer at low temperatures. Furthermore, symmetry-protected nodal lines near the Fermi level substantially enhance interband transitions, thereby increasing the resistivity-like damping. Our findings predict that the damping can be reduced to arbitrarily low values by **decreasing the electronic scattering rate**. Additionally, we find a pronounced damping anisotropy when the magnetization is rotated from an out-of-plane to an in-plane orientation. These unique features are generally applicable to other 2D vdW materials with mirror symmetry, such as Fe_3GeTe_2 and 2H-FeTe_2 .

Theoretical methods.—We utilize the **torque-correlation model** [40–42] to calculate the Gilbert damping of 2D vdW FM metals based upon first-principles electronic structure. This model captures the dynamical dissipation of magnetization due to spin-orbit coupling (SOC), corresponding to the dominant

contribution to the **intrinsic Gilbert damping** α [43]. Explicitly, α is given by

$$\alpha = \frac{\pi g \mu_B}{M_s} \sum_{n,m} \int \frac{d^3 k}{(2\pi)^3} |\Gamma_{mn}^-(\mathbf{k})|^2 \int d\epsilon \eta(\epsilon) A_{n\mathbf{k}}(\epsilon) A_{m\mathbf{k}}(\epsilon). \quad (1)$$

Here g , μ_B and M_s are the Landé factor, the Bohr magneton and the saturation magnetization, respectively. While indices n and m run over all energy bands, the energy derivative of the Fermi-Dirac distribution $\eta(\epsilon) \equiv -\partial f / \partial \epsilon$ ensures that the damping is predominantly influenced by electronic states near the Fermi surface. The matrix element $\Gamma_{mn}^-(\mathbf{k}) = \langle \Psi_{m\mathbf{k}} | [\sigma^-, H_{\text{soc}}] | \Psi_{n\mathbf{k}} \rangle$ characterizes the torque on electron spin stemming from SOC. The **spectral function** $A_{n\mathbf{k}}(\epsilon)$ is modeled as a **Lorentzian centered at the band energy** $\epsilon_{n\mathbf{k}}$, with its width determined by the electronic scattering rate γ [41, 42]. Prior studies have shown that intraband ($n = m$) transitions yield a **conductivity-like damping component**, which is proportional to the electronic relaxation time, $\tau = \hbar / 2\gamma$ [40, 41], while **interband ($n \neq m$) transitions contribute to a resistivity-like damping component that scales monotonically with increasing γ** [44]. To explicitly evaluate these intraband and interband transitions in Eq. (1), we have developed a Wannier interpolation method to perform the integral of $|\Gamma_{mn}^-(\mathbf{k})|^2$ based on the first-principles electronic structures. The self-consistent band structures are calculated using Quantum ESPRESSO [45], and the Wannier functions are constructed using WANNIER90 [46]. Here, we focus on the intrinsic Gilbert damping induced by SOC, while we do not consider extrinsic mechanisms such as two-magnon scattering [47, 48] or the Stoner damping mediated by electron-electron interaction in strongly correlated systems [49–51].

Intraband transitions.— We choose Fe_3GaTe_2 as a typical example to illustrate the unique characteristics of Gilbert damping in 2D vdW ferromagnets. The lattice structure of a single layer Fe_3GaTe_2 , referred to as one quintuple-layer (1QL), is depicted in Fig. 1(a). This lattice, belonging to the D_{3h} point group, maintains mirror symmetry about the blue plane only when the magnetization is perpendicular to the atomic plane [15]. The Gilbert damping α of 1QL Fe_3GaTe_2 , calculated using Eq. (1), is plotted in Fig. 1(b) **as a function of electronic scattering rate γ** . With out-of-plane magnetization ($\mathbf{m} \parallel \langle 0001 \rangle$), α increases monotonically with increasing γ . Detailed analysis shows that the intraband transitions are completely absent, leading to vanishing conductivity-like behavior. As a result, α is lower than 10^3 at small scattering rate $\gamma < 10$ meV. In contrast, for in-plane magnetization ($\mathbf{m} \parallel \langle 2\bar{1}\bar{1}0 \rangle$), α exhibits a non-monotonic dependence on γ , with both conductivity-like and resistivity-like terms contributing.

The ultralow damping due to the absence of conductivity-like damping with out-of-plane magnetiza-

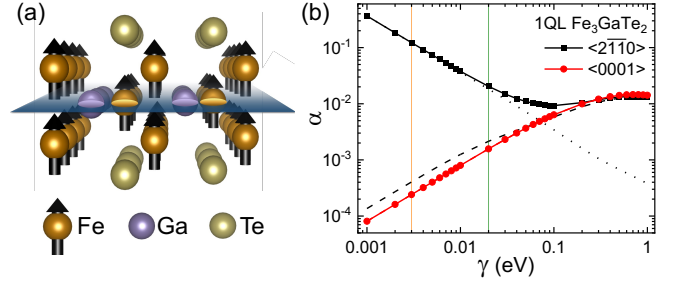


FIG. 1. (a) Atomic structure of 1QL Fe_3GaTe_2 . The magnetic moments on the Fe atoms are indicated by black arrows. The blue plane in the middle of the layer highlights the mirror symmetry of the lattice structure. (b) Gilbert damping as a function of the electronic scattering rate for 1QL Fe_3GaTe_2 with the out-of-plane (red solid circles) and in-plane magnetization (black solid squares). The dotted and dashed lines represent the contributions from intraband and interband transitions, respectively. The orange and green vertical lines denote two characteristic scattering rates, $\gamma = 3$ meV and 20 meV, which are used to calculate the damping in detail.

tion can be explained through a symmetry analysis of the electronic states. We define $\mathcal{M}_z$ as the mirror operator [52] about the central plane of the 1QL Fe_3GaTe_2 . For $\mathbf{m} \parallel \langle 0001 \rangle$, the system is mirror-symmetric, and thus $\mathcal{M}_z$ commutes with the total Hamiltonian, as well as the SOC Hamiltonian H_{soc} [52]. Consequently, each Bloch state $\Psi_{n\mathbf{k}}$ of the system is an eigenfunction of $\mathcal{M}_z$ with eigenvalues $a_{n\mathbf{k}} = -i$ (symmetric) or $+i$ (antisymmetric). These mirror symmetry properties of Bloch states for 1QL Fe_3GaTe_2 are explicitly calculated using the **IRVSP code** [53], as shown in Fig. 2(a), where red and blue lines represent symmetric and antisymmetric energy bands with out-of-plane magnetization. Since $\mathcal{M}_z$ anti-commutes with σ^- , we have $\{\mathcal{M}_z, [\sigma^-, H_{\text{soc}}]\} = 0$. Considering the mirror symmetry properties of Bloch states, we find

$$\Gamma_{mn}^-(\mathbf{k}) = -a_{m\mathbf{k}}^* a_{n\mathbf{k}} \Gamma_{mn}^-(\mathbf{k}). \quad (2)$$

For intraband transitions ($m = n$), $|a_{n\mathbf{k}}|^2 = 1$ always holds, leading to $\Gamma_{nn}^-(\mathbf{k}) = -\Gamma_{nn}^-(\mathbf{k})$. Thus we have $\Gamma_{nn}^-(\mathbf{k}) = 0$ indicating that the intraband transitions are completely forbidden, regardless of the symmetry of the Bloch states. Consequently, the conductivity-like damping vanishes and α increases monotonically with γ .

Alternatively, the symmetry-forbidden intraband transitions can be intuitively understood by examining the energy dissipation during magnetization dynamics within Kamberský’s breathing Fermi surface model [40, 44]. When the magnetization is out-of-plane at equilibrium, the transverse (in-plane) components of the spin angular momentum $\Delta \mathbf{S}$ relax, as illustrated in Fig. 2(b). The energy changes with the spin variation $\Delta \mathbf{S}$ through SOC, i.e., $\Delta E = \Delta H_{\text{soc}} \propto \mathbf{L} \cdot \Delta \mathbf{S}$. Due to the mirror symmetry, the orbital angular momentum $\mathbf{L}$ must have

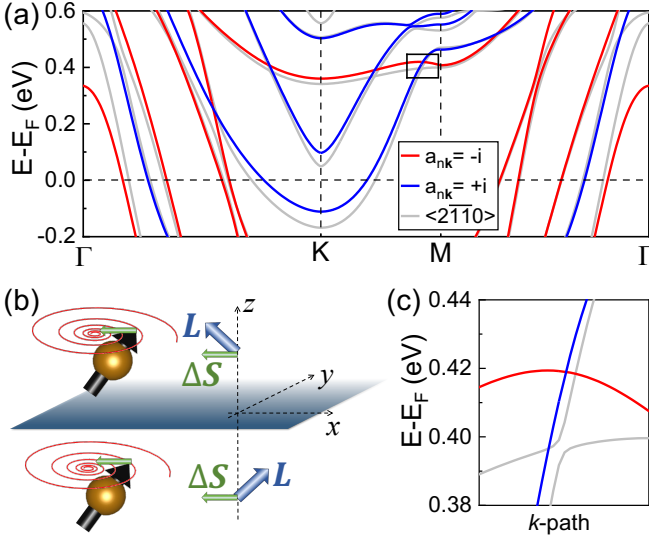


FIG. 2. (a) Calculated band structure of 1QL Fe_3GaTe_2 with out-of-plane magnetization. The red and blue lines denote mirror-symmetric ($a_{n\mathbf{k}} = -i$) and antisymmetric ($a_{n\mathbf{k}} = +i$) bands, respectively. Bands calculated with in-plane magnetization are presented in gray. (b) Schematic illustration of energy dissipation during magnetization dynamics in a mirror-symmetric system. (c) Enlarged view of a typical symmetry-protected band crossing, as indicated by the rectangle in (a).

the same out-of-plane component but opposite in-plane components on either side of the mirror plane, namely $L_z(z) = L_z(-z)$ and $L_{x,y}(z) = -L_{x,y}(-z)$. Therefore, after integrating over the entire space, the band energy remains unchanged with the in-plane $\Delta\mathbf{S}$, i.e., $\Delta E = 0$. This implies that the breathing Fermi surface during the magnetization dynamics does not excite the system out of equilibrium, and the intraband transitions are unnecessary to bring it back to equilibrium.

To further confirm that the absent intraband transitions (conductivity-like damping) is indeed a result of symmetry, we intentionally break the mirror symmetry in different ways. First, we rotate the magnetization of 1QL Fe_3GaTe_2 away from the surface normal. This rotation means the Bloch states are no longer eigenstates of $\mathcal{M}_z$. The calculated energy bands for in-plane magnetization are plotted with gray lines in Fig. 2(a), where the hybridization of the red and blue bands forms anticrossing gaps with a typical example magnified in Fig. 2(c). Such band hybridization facilitates intraband transitions, leading to the emergence of the conductivity-like damping due to intraband transitions, corresponding to the black squares in Fig. 1(b). We select two scattering rates, $\gamma = 3$ meV and 20 meV, as marked by the vertical lines in Fig. 1(b), and calculate damping as a function of the tilting angle θ . As shown in Fig. 3(a), the calculated α increases monotonically with θ and tends to saturate at $\theta > 60^\circ$. The saturation suggests that the energy bands

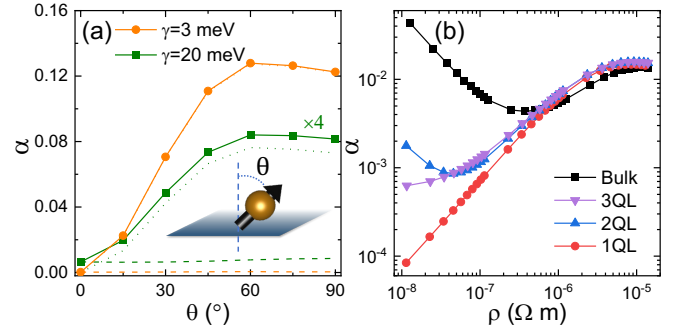


FIG. 3. (a) Gilbert damping for 1QL Fe_3GaTe_2 as a function of the tilting angle θ of the magnetization away from the surface normal, for two scattering rates $\gamma = 3$ meV and 20 meV. The intraband and interband contributions to the damping are represented by dotted and dashed curves, respectively. (b) Gilbert damping for Fe_3GaTe_2 with varying stacking layers plotted as a function of electric resistivity.

hybridization is strong enough to no longer be treated as a perturbation. The enhanced damping is dominated by the intraband transitions (the dotted lines) for both γ values, while the interband contributions (the dashed lines) remain nearly unchanged. We conducted an analysis on Fe_3GeTe_2 , whose lattice structure is identical to Fe_3GaTe_2 , except for the substitution of Ga atoms with Ge. For 1 QL Fe_3GeTe_2 with perpendicular magnetization, conductivity-like damping vanishes, resulting in a monotonic increase of the calculated α with increasing γ . Rotating the magnetization leads to recovery of intraband contributions, thus yielding non-monotonic behavior, which is qualitatively the same as Fe_3GaTe_2 ; see Sec. 1 in Supplemental Material [54].

The second approach involves varying the number of stacking layers of Fe_3GaTe_2 . By analyzing the lattice structures of 1QL, 2QL and 3QL Fe_3GaTe_2 , we find that mirror symmetry is preserved for odd-number stacking layers but broken in the 2QL system. The calculated damping of Fe_3GaTe_2 with different numbers of stacking layers is shown in Fig. 3(b), plotted as a function of the simultaneously calculated **electrical resistivity** ρ . The resistivity is calculated using the Boltzmann equation within the relaxation time approximation, where ρ is **proportional to the scattering rate** $\gamma = \hbar/2\tau$. The calculated α for 1QL and 3QL both monotonically increase with ρ , indicating the forbidden intraband transitions, as expected due to mirror symmetry. In contrast, α of 2QL Fe_3GaTe_2 exhibits a nonmonotonic dependence on ρ with noticeable conductivity-like damping arising from intraband transitions at low ρ . The calculated α for bulk Fe_3GaTe_2 , shown as the black squares in Fig. 3(b), contains a much larger conductivity-like component. This is because the mirror symmetry is well defined only for the Bloch states with $k_z = 0$ and those k_z located at the boundary plane of the first Brillouin zone [52]. Other electronic states be-

tween these special planes are not eigenstates of the mirror operator and intraband transitions are not prohibited by mirror symmetry. Compared with the bulk case, the significantly reduced damping in Fe_3GaTe_2 thin films at low ρ manifests the strong suppression of the intraband transitions and thus the conductivity-like behavior.

We further performed calculations for 2D monolayer FeTe_2 in two distinct structural phases: the trigonal prismatic 2H phase and the octahedral 1T phase. The mirror-symmetric 2H- FeTe_2 with out-of-plane magnetization exhibits monotonic damping. In contrast, the damping in 1T- FeTe_2 , which lacks mirror symmetry, behaves non-monotonically and is independent of magnetization orientation; see Sec. 2 in Supplemental Material [54]. These additional calculations indicate that the symmetry-forbidden intraband transition is a general feature of any mirror-symmetric 2D vdW ferromagnets.

Interband transitions.—Although intraband transitions are prohibited by mirror symmetry, Eq. (2) indicates that interband transitions are allowed between Bloch states with contrasting symmetry characteristics, i.e., $a_{m\mathbf{k}} \neq a_{n\mathbf{k}}$. In fact, the mirror symmetry ensures the presence of nodal lines in the Brillouin zone, which are formed by the intersections of mirror-symmetric and mirror-antisymmetric bands [55, 56]. According to Eq. (1), the interband contribution to damping depends on the overlap of the broadened energy bands at the Fermi energy E_F . As schematically illustrated in Fig. 4(a), the interband contribution from the two intersecting bands is maximized when the intersection point aligns with the Fermi level, i.e., $E_F = E_3$. Consequently, pronounced interband transitions are anticipated near band crossings or nodal lines close to E_F . In the case of 1QL Fe_3GaTe_2 , the nodal lines in the 2D Brillouin zone are located several hundred meV above E_F , as shown in Fig. 4(b). This accounts for the very low interband damping of 1QL Fe_3GaTe_2 with out-of-plane magnetization, particularly at low γ .

The enhancement of damping by nodal lines is confirmed by elevating E_F in 1QL Fe_3GaTe_2 . Two scattering rates, $\gamma = 3$ meV and 20 meV, were selected, and the calculated damping is plotted in Fig. 4(c) as a function of energy. Near the original E_F , α is minimal and insensitive to energy shifts since the intraband transitions are forbidden and the nodal lines are far above E_F . For $E - E_F > 200$ meV, as energy approaches the nodal lines, α increases rapidly for both γ values due to significantly enhanced interband transitions. We further investigate the scattering rate dependence of α at various energies. As shown in Fig. 4(d), the typical monotonic resistivity-like damping is observed at $E - E_F = 100$ and 200 meV, where the nodal lines are not significantly involved. When scattering occurs at $E - E_F = 300$ and 400 meV, damping becomes substantial even at very low γ . This is because the interband transitions always occur at the crossing points, irrespective of broadening. Thus,

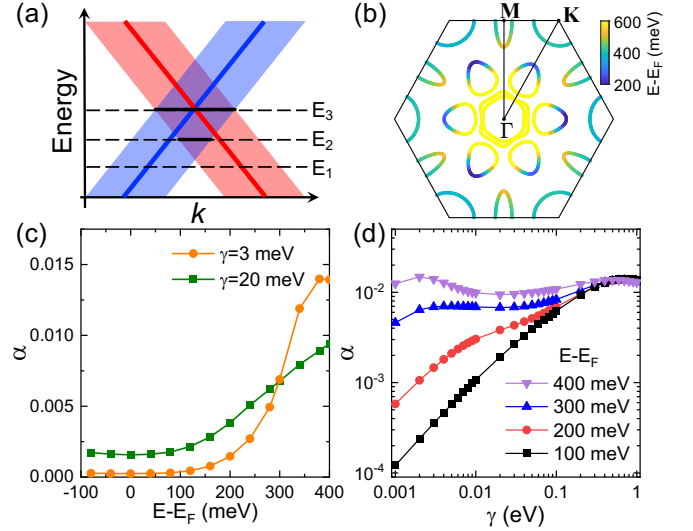


FIG. 4. (a) Schematic representation of band overlap near their crossing point, with shadows indicating the broadening. Thick black lines show the extent of overlap at various isoenergetic surfaces. (b) Calculated nodal lines for 1QL Fe_3GaTe_2 with out-of-plane magnetization in the 2D Brillouin zone. (c) Calculated α as a function of energy at two scattering rates. (d) Calculated α as a function of scattering rate at various energies.

the damping displays a very weak dependence on γ . It also implies that the common assumption $\alpha_{\text{interband}} \propto \rho$ does not apply in these instances.

The interband transitions also explain the higher α calculated for 3QL Fe_3GaTe_2 compared to 1QL, as shown in Fig 3(b). With mirror symmetry maintained, we analyze the band structure for 3QL (Sec. 3 in Supplemental Material [54]). The tripling of atoms in the unit cell results in a greater number of bands in 3QL compared to 1QL. Consequently, the denser bands lead to a larger band overlap under the same broadening γ , culminating in a substantially larger interband contribution to damping for 3QL than for 1QL.

Discussions and conclusions.—The predicted features in Gilbert damping of mirror-symmetric vdW ferromagnets can be directly measured in experiments. The symmetry-forbidden intraband transitions with out-of-plane magnetization results in a low damping. By rotating the magnetization using an external field, the mirror symmetry is broken, leading to the recovery of the intraband (conductivity-like) damping. Consequently, the Gilbert damping can be controlled by adjusting the magnetization orientation without altering the scattering rate, which is particularly significant in clean systems. The calculated damping anisotropy between out-of-plane and in-plane magnetization for Fe_3GaTe_2 is depicted in Fig. 5. For both 1QL and 3QL, a substantial anisotropy ratio exceeding $10^4\%$ is observed at small γ , and it exceeds 200% for bulk. This damping anisotropy is ex-

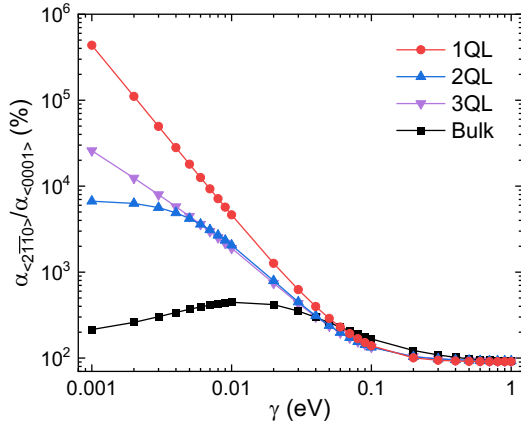


FIG. 5. Calculated damping anisotropy between out-of-plane and in-plane magnetization for Fe_3GaTe_2 .

pected to be observable using ferromagnetic resonance or time-resolved magneto-optical Kerr effect. Additionally, the monotonic (non-monotonic) temperature-dependent damping for odd (even) layers can also be experimentally measured, as suggested by Fig. 3(b).

In summary, our investigation into the Gilbert damping of 2D vdW FM metals has uncovered the crucial role of mirror symmetry in governing damping behavior. In 1QL Fe_3GaTe_2 , the conductivity-like damping vanishes under out-of-plane magnetization due to symmetry-forbidden intraband transitions. This results in an ultralow Gilbert damping at small scattering rate. However, this symmetry can be broken either by tilting the magnetization or by varying the number of layers, which reinstates the intraband contribution and substantially enhances damping. Furthermore, symmetry-protected crossings between mirror-symmetric and antisymmetric bands create nodal lines where the interband transitions are pronounced, leading to a significant damping enhancement. Voltage control offers a promising avenue for tuning the damping by shifting the Fermi level relative to these nodal lines. These nodal lines and associated topological electronic states may have interesting interactions with topological magnons in vdW FM systems [57–60]. Our findings not only elucidate the microscopic origins of the observed monotonic temperature dependence of Gilbert damping but also predict a universal mechanism to achieve the ultralow Gilbert damping in 2D vdW FM materials, with no lower limit.

This work was supported by the National Key Research and Development Program of China (2024YFA1408500), the National Natural Science Foundation of China (Grants No. 12374101 and No. 12174028), the Open Fund of the State Key Laboratory of Spintronics Devices and Technologies (Grant No. SPL-2408) and the Open Project of Guangdong Provincial Key Laboratory of Magnetoelectric Physics and Devices (2022B1212010008).

* yiliu42@shu.edu.cn

† yuanz@fudan.edu.cn

- [1] B. Huang, G. Clark, E. Navarro-Moratalla, D. R. Klein, R. Cheng, K. L. Seyler, D. Zhong, E. Schmidgall, M. A. McGuire, D. H. Cobden, W. Yao, D. Xiao, P. Jarillo-Herrero and X. Xu, Layer-dependent ferromagnetism in a van der Waals crystal down to the monolayer limit, *Nature* **546**, 270 (2017).
- [2] C. Gong, L. Li, Z. Li, H. Ji, A. Stern, Y. Xia, T. Cao, W. Bao, C. Wang, Y. Wang, Z. Q. Qiu, R. J. Cava, S. G. Louie, J. Xia and X. Zhang, Discovery of intrinsic ferromagnetism in two-dimensional van der Waals crystals, *Nature* **546**, 265 (2017).
- [3] Z. Fei, B. Huang, P. Malinowski, W. Wang, T. Song, J. Sanchez, W. Yao, D. Xiao, X. Zhu, A. F. May, W. Wu, D. H. Cobden, J.-H. Chu and X. Xu, Two-dimensional itinerant ferromagnetism in atomically thin Fe_3GeTe_2 , *Nat. Mater.* **17**, 778 (2018).
- [4] Y. Deng, Y. Yu, Y. Song, J. Zhang, N. Z. Wang, Z. Sun, Y. Yi, Y. Z. Wu, S. Wu, J. Zhu, J. Wang, X. H. Chen, and Y. Zhang, Gate-tunable room-temperature ferromagnetism in two-dimensional Fe_3GeTe_2 , *Nature* **563**, 94 (2018).
- [5] C. Gong, X. Zhang, Two-dimensional magnetic crystals and emergent heterostructure devices, *Science* **363**, eaav4450 (2019).
- [6] B. Dieny, I. L. Prejbeanu, K. Garello, P. Gambardella, P. Freitas, R. Lehnndorff, W. Raberg, U. Ebels, S. O. Demokritov, J. Akerman, A. Deac, P. Pirro, C. Adelman, A. Anane, A. V. Chumak, A. Hirohata, S. Mangin, S. O. Valenzuela, M. C. Onbasli, M. d’Aquino, G. Prenat, G. Finocchio, L. Lopez-Diaz, R. Chantrell, O. Chubykalo-Fesenko and P. Bortolotti, Opportunities and challenges for spintronics in the microelectronics industry, *Nat. Electronics* **3**, 446 (2020).
- [7] K. S. Burch, D. Mandrus, and J.-G. Park, Magnetism in two-dimensional van der Waals materials, *Nature* **563**, 47 (2018).
- [8] M. Blei, J. L. Lado, Q. Song, D. Dey, O. Erten, V. Pardo, R. Comin, S. Tongay, and A. S. Botana, Synthesis, engineering, and theory of 2D van der Waals magnets, *Appl. Phys. Rev.* **8**, 021301 (2021).
- [9] X. Wang, J. Tang, X. Xia, C. He, J. Zhang, Y. Liu, C. Wan, C. Fang, C. Guo, W. Yang, Y. Guang, X. Zhang, H. Xu, J. Wei, M. Liao, X. Lu, J. Feng, X. Li, Y. Peng, H. Wei, R. Yang, D. Shi, X. Zhang, Z. Han, Z. Zhang, G. Zhang, G. Yu, X. Han, Current-driven magnetization switching in a van der Waals ferromagnet Fe_3GeTe_2 , *Sci. Adv.* **5**, eaaw8904 (2019).
- [10] M. Alghamdi, M. Lohmann, J. Li, P. R. Jothi, Q. Shao, M. Aldosary, T. Su, B. P. T. Fokwa, and J. Shi, Highly efficient spin-orbit torque and switching of layered ferromagnet Fe_3GeTe_2 , *Nano Lett.* **19**, 4400 (2019).
- [11] H. Wang, H. Wu, J. Zhang, Y. Liu, D. Chen, C. Pandey, J. Yin, D. Wei, N. Lei, S. Shi, H. Lu, P. Li, A. Fert, K. L. Wang, T. Nie and W. Zhao, Room temperature energy-efficient spin-orbit torque switching in two-dimensional van der Waals Fe_3GeTe_2 induced by topological insulators, *Nat. Commun.* **14**, 5173 (2023).
- [12] Z. Wang, D. Sapkota, T. Taniguchi, K. Watanabe, D. Mandrus, and A. F. Morpurgo, Tunneling spin valves

- based on $\text{Fe}_3\text{GeTe}_2/\text{hBN}/\text{Fe}_3\text{GeTe}_2$ van der Waals heterostructures, *Nano Lett.* **18**, 4303 (2018).
- [13] Z.-A. Wang, W. Xue, F. Yan, W. Zhu, Y. Liu, X. Zhang, Z. Wei, K. Chang, Z. Yuan, and K. Wang, Selectively controlled ferromagnets by electric fields in van der Waals ferromagnetic heterojunctions, *Nano Lett.* **23**, 710 (2023).
 - [14] J. Xiong, J. Xie, B. Cheng, Y. Dai, X. Cui, L. Wang, Z. Liu, J. Zhou, N. Wang, X. Xu, X. Chen, S.-W. Cheong, S.-J. Liang and F. Miao, Electrical switching of Ising-superconducting nonreciprocity for quantum neuronal transistor, *Nat. Commun.* **15**, 4953 (2024).
 - [15] G. Zhang, F. Guo, H. Wu, X. Wen, L. Yang, W. Jin, W. Zhang and H. Chang, Above-room-temperature strong intrinsic ferromagnetism in 2D van der Waals Fe_3GaTe_2 with large perpendicular magnetic anisotropy, *Nat. Commun.* **13**, 5067 (2022).
 - [16] H. Pan, C. Zhang, J. Shi, X. Hu, N. Wang, L. An, R. Duan, P. Deb, Z. Liu, and W. Gao, Room-temperature lateral spin valve in graphene/ Fe_3GaTe_2 van der Waals heterostructures, *ACS Mater. Lett.* **5**, 2226 (2023).
 - [17] W. Jin, G. Zhang, H. Wu, L. Yang, W. Zhang and H. Chang, Room-temperature spin-valve devices based on $\text{Fe}_3\text{GaTe}_2/\text{MoS}_2/\text{Fe}_3\text{GaTe}_2$ 2D van der Waals heterojunctions, *Nanoscale* **15**, 5371 (2023).
 - [18] Y. Deng, M. Wang, Z. Xiang, K. Zhu, T. Hu, L. Lu, Y. Wang, Y. Ma, B. Lei, X. Chen, Room-temperature highly efficient nonvolatile magnetization switching by current in van der Waals Fe_3GaTe_2 devices, *Nano Lett.* **24**, 9302 (2024).
 - [19] Y. Zhang, X. Ren, R. Liu, Z. Chen, X. Wu, J. Pang, W. Wang, G. Lan, K. Watanabe, T. Taniguchi, Y. Shi, G. Yu, Q. Shao, Robust field-free switching using large unconventional spin-orbit torque in an all-van der Waals heterostructure, *Adv. Mater.* **36**, 2406464 (2024).
 - [20] C. Tang, L. Alahmed, M. Mahdi, Y. Xiong, J. Inman, N. J. McLaughlin, C. Zollitsch, T. H. Kim, C. R. Du, H. Kurebayashi, E. J. G. Santos, W. Zhang, P. Li, W. Jin, Spin dynamics in van der Waals magnetic systems, *Phys. Rep.* **1032**, 1 (2023).
 - [21] T. L. Gilbert, A Phenomenological Theory of Damping in Ferromagnetic Materials, *IEEE Trans. Magn.* **40**, 3443 (2004).
 - [22] A. Brataas, A. D. Kent, H. Ohno, Current-induced torques in magnetic materials, *Nat. Mater.* **11**, 372 (2012).
 - [23] J. Z. Sun, Spin-current interaction with a monodomain magnetic body: A model study, *Phys. Rev. B* **62**, 570 (2000).
 - [24] G. Tatara, H. Kohno, J. Shibata, Microscopic approach to current-driven domain wall dynamics, *Phys. Rep.* **468**, 213 (2008).
 - [25] Z. Wang, M. Guo, H.-A. Zhou, L. Zhao, T. Xu, R. Tomasello, H. Bai, Y. Dong, S.-G. Je, W. Chao, H.-S. Han, S. Lee, K.-S. Lee, Y. Yao, W. Han, C. Song, H. Wu, M. Carpentieri, G. Finocchio, M.-Y. Im, S.-Z. Lin and W. Jiang, Thermal generation, manipulation and thermoelectric detection of skyrmions, *Nat Electron* **3**, 672 (2020).
 - [26] M. A. W. Schoen, D. Thonig, M. L. Schneider, T. J. Silva, H. T. Nembach, O. Eriksson, O. Karis, and J. M. Shaw, Ultra-low magnetic damping of a metallic ferromagnet, *Nat. Phys.* **12**, 839 (2016).
 - [27] A. J. Lee, J. T. Brangham, Y. Cheng, S. P. White, W. T. Ruane, B. D. Esser, D. W. McComb, P. C. Hammel, and F. Yang, Metallic ferromagnetic films with magnetic damping under 1.4×10^{-3} , *Nat. Commun.* **8**, 234 (2017).
 - [28] Y. Wei, W. Zhang, B. Lv, X. Xu, S. Xi, and Z. Ma, Ultralow magnetic damping of a common metallic ferromagnetic film, *Sci. Adv.* **7**, eabc5053 (2021).
 - [29] M. Arora, E. K. Delczeg-Czirjak, G. Riley, T. J. Silva, H. T. Nembach, O. Eriksson, and J. M. Shaw, Magnetic damping in polycrystalline thin-film Fe-V alloys, *Phys. Rev. Applied* **15**, 054031 (2021).
 - [30] C. Guillemard, S. Petit-Watelot, L. Pasquier, D. Pierre, J. Ghanbaja, J.-C. Rojas-Sánchez, A. Bataille, J. Rault, P. Le Fèvre, F. Bertran, and S. Andrieu, Ultralow magnetic damping in Co_2Mn -based Heusler compounds: Promising materials for spintronics, *Phys. Rev. Appl.* **11**, 064009 (2019).
 - [31] Z. Zhang, M. Cheng, Z. Yu, Z. Zou, Y. Liu, J. Shi, Z. Lu, and R. Xiong, Ultralow Gilbert damping in CrO_2 epitaxial films, *Phys. Rev. B* **102**, 014454 (2020).
 - [32] S. M. Bhagat and P. Lubitz, Temperature variation of ferromagnetic relaxation in the 3d transition metals, *Phys. Rev. B* **10**, 179 (1974).
 - [33] B. Heinrich, D. J. Meredith, and J. F. Cochran, Wave number and temperature-dependent Landau-Lifshitz damping in nickel, *J. Appl. Phys.* **50**, 7726 (1979).
 - [34] B. Khodadadi, A. Rai, A. Sapkota, A. Srivastava, B. Nepal, Y. Lim, D. A. Smith, C. Memes, S. Budhathoki, A. J. Hauser, M. Gao, J.-F. Li, D. D. Viehland, Z. Jiang, J. J. Heremans, P. V. Balachandran, T. Mewes, and S. Emori, Conductivitylike Gilbert damping due to intraband scattering in epitaxial iron, *Phys. Rev. Lett.* **124**, 157201 (2020).
 - [35] D. Zhang, D. Huang, R. J. Wu, D. Lattery, J. Liu, X. Wang, D. B. Gopman, K. A. Mkhyon, J.-P. Wang, and X. Wang, Low Gilbert damping and high thermal stability of Ru-seeded L_{10} -phase FePd perpendicular magnetic thin films at elevated temperatures, *Appl. Phys. Lett.* **117**, 082405 (2020).
 - [36] W. K. Peria, M. B. Katz, J.-P. Wang, P. A. Crowell and D. B. Gopman, Low Gilbert damping and high perpendicular magnetic anisotropy in an Ir-coupled L_{10} -FePd-based synthetic antiferromagnet, *Sci. Rep.* **14**, 13290 (2024).
 - [37] J. Zhang, Z. Wang, Z. Li, T. Li, S. Liu, J. Zhang, R.-J. Zhang, Q. Jin, Z. Shi, Y. Liu, Z. Sheng, and Z. Zhang, Sub-THz high spin precession frequency in van der Waals ferromagnet Fe_3GaTe_2 , *Nano Lett.* **24**, 12204 (2024).
 - [38] P. T. Yang, R. X. Liu, Z. Yuan and Y. Liu, Magnetic damping anisotropy in the two-dimensional van der Waals material Fe_3GeTe_2 from first principles, *Phys. Rev. B* **106**, 134409 (2022).
 - [39] L. Alahmed, B. Nepal, J. Macy, W. Zheng, B. Casas, A. Sapkota, N. Jones, A. R. Mazza, M. Brahlek, W. Jin, M. Mahjouri-Samani, S. S.-L. Zhang, C. Mewes, L. Balicas, T. Mewes, and P. Li, Magnetism and spin dynamics in room-temperature van der Waals magnet Fe_3GeTe_2 , *2D Mater.* **8**, 045030 (2021).
 - [40] V. Kamberský, On the Landau-Lifshitz relaxation in ferromagnetic metals, *Can. J. Phys.* **48**, 2906 (1970).
 - [41] K. Gilmore, Y. U. Idzerda, and M. D. Stiles, Identification of the dominant precession-damping mechanism in Fe, Co, and Ni by first-principles calculations, *Phys. Rev. Lett.* **99**, 027204 (2007).
 - [42] K. Gilmore, Y. U. Idzerda, and M. D. Stiles, Spin-orbit

- precession damping in transition metal ferromagnets, *J. Appl. Phys.* **103**, 07D303 (2008).
- [43] B. Heinrich, Spin relaxation in magnetic metallic layers and multilayers, in *Ultrathin Magnetic Structures III*, edited by J. A. C. Bland and B. Heinrich, (Springer-Verlag, Berlin, 2005).
- [44] V. Kamberský, On ferromagnetic resonance damping in metals, *Czech. J. Phys. B* **28**, 1366 (1976).
- [45] P. Giannozzi *et al.*, Advanced capabilities for materials modelling with Quantum ESPRESSO, *J. Phys.: Condens. Matter* **29**, 465901 (2017).
- [46] G. Pizzi *et al.*, Wannier90 as a community code: new features and applications, *J. Phys.: Condens. Matter* **32**, 165902 (2020).
- [47] R. Arias and D. L. Mills, Extrinsic contributions to the ferromagnetic resonance response of ultrathin films, *Phys. Rev. B* **60**, 7395 (1999).
- [48] K. Lenz, H. Wende, W. Kuch, K. Baberschke, K. Nagy and A. Jánosy, Two-magnon scattering and viscous Gilbert damping in ultrathin ferromagnets, *Phys. Rev. B* **73**, 144424 (2006).
- [49] M. Sayad, R. Rausch, and M. Potthoff, Relaxation of a classical spin coupled to a strongly correlated electron system, *Phys. Rev. B* **117**, 127201 (2016).
- [50] E. G. C. P. van Loon and H. U. R. Strand, Larmor precession in strongly correlated itinerant electron systems, *Commun. Phys.* **6**, 289 (2023).
- [51] A. Szilva, Y. Kvashnin, E. A. Stepanov, L. Nordström, O. Eriksson, A. I. Lichtenstein, M. I. Katsnelson, Quantitative theory of magnetic interactions in solids, *Rev. Mod. Phys.* **95**, 035004 (2023).
- [52] F. Zhang, C. L. Kane, and E. J. Mele, Topological mirror superconductivity, *Phys. Rev. Lett.* **111**, 056403 (2013).
- [53] J. Gao, Q. Wu, C. Persson, Z. Wang, Irvsp: To obtain irreducible representations of electronic states in the VASP, *Comput. Phys. Commun.* **261**, 107760 (2021).
- [54] See Supplemental Material at [URL] for calculated Gilbert damping of Fe_3GeTe_2 and FeTe_2 , and band structure of 3QL Fe_3GaTe_2 .
- [55] S.-Y. Yang, H. Yang, E. Derunova, S. S. P. Parkin, B. Yan, and M. N. Ali, Symmetry demanded topological nodal-line materials, *Adv. Phys. X* **3**, 1414631 (2018).
- [56] F. L. Zeng, Z. Y. Ren, Y. Li, J. Y. Zeng, M. W. Jia, J. Miao, A. Hoffmann, W. Zhang, Y. Z. Wu, and Z. Yuan, Intrinsic mechanism for anisotropic magnetoresistance and experimental confirmation in $\text{Co}_x\text{Fe}_{1-x}$ single-crystal films, *Phys. Rev. Lett.* **125**, 097201 (2020).
- [57] L. Chen, J.-H. Chung, B. Gao, T. Chen, M. B. Stone, A. I. Kolesnikov, Q. Huang, and P. Dai, Topological spin excitations in honeycomb ferromagnet CrI_3 , *Phys. Rev. X* **8**, 041028 (2018).
- [58] E. Aguilera, R. Jaeschke-Ubiergo, N. Vidal-Silva, L. E. F. Foa Torres, and A. S. Nunez, Topological magnonics in the two-dimensional van der Waals magnet CrI_3 , *Phys. Rev. B* **102**, 024409 (2020).
- [59] R. Jaeschke-Ubiergo, E. Suárez Morell, and A. S. Nunez, Theory of magnetism in the van der Waals magnet CrI_3 , *Phys. Rev. B* **103**, 174410 (2021).
- [60] Z.-X. Shen, X. Bo, K. Cao, X. Wan, and L. He, Magnetic ground state and electron-doping tuning of Curie temperature in Fe_3GeTe_2 : First-principles studies, *Phys. Rev. B* **103**, 085102 (2021).